NATIONAL UNIVERSITY OF SCIENCES & TECHNOLOGY

Machine Learning (ESP)

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TRANSFORMER BASED GRAMMAR AID

1. **Novelty** Most grammar correction tools like Grammarly are commercial and closed source. In this project, I'm building an open-source AI grammar correction and feedback system using pre-trained Transformer models. The new part (novelty) is that it doesn't just correct mistakes, it also explains what was wrong, which makes it useful for students and learners. It will run completely on Google Colab (free), so no paid tools or training data are required.

2. **Abstract** The purpose of this project is to develop an AI tool that can correct English grammar mistakes and also give short feedback on what was changed. It uses a Transformer-based pre-trained model to automatically find and fix grammatical errors in sentences. The model will be implemented in Google Colab using the Hugging Face Transformers library, and a simple user interface will be made using Gradio. The project shows how machine learning can be used to improve writing skills and make grammar learning easier for students.

3. **Introduction** Grammar correction is one of the main applications of natural language processing (NLP). Traditional rule-based systems often fail to handle complex sentence structures or modern language usage. With the help of Transformer-based models like T5 and BART, it's now possible to automatically correct grammar with high accuracy. In this project, I plan to build a grammar correction and writing feedback assistant using a pre-trained Transformer model. It will take any English sentence, correct it, and give short feedback such as "article missing" or "verb form changed." This can help students improve their writing and understand grammar errors in real time.

4. Related Work

a) Gramformer (Prithivida, 2021): Transformer-based grammar correction model trained on English data.

b) GECToR (Omelianchuk et al., 2020): A Transformer model designed for fast and accurate grammar correction.

c) T5 (Raffel et al., 2020): General-purpose Transformer that can perform multiple text-to-text tasks.

d) Tools like Grammarly and Ginger also correct grammar but are not open or customizable.

5. Methodology

Step	Description
1. Model	• Use the pre-trained grammar correction model: prithivida/grammar_error_correcter_v1 (based on T5).
2. Text Processing	• User writes any English sentence.• Model predicts the grammatically correct version of the sentence.
3. Feedback Generation	• Compare original and corrected sentences using Python library difflib. • Highlight differences and explain the type of grammar error (verb form, article, tense, etc.).
4. Interface	• Use Gradio to make a simple web interface • Components: – Input text box – Corrected output – Feedback section
5. Tools and Environment	• Platform: Google Colab (Free) • Python libraries: – transformers – torch – gradio – difflib

6. Results (Expected)

a) The model will correct most English grammar mistakes accurately.

b) It will display both the corrected sentence and a short explanation.

c) The tool should run smoothly in Colab without extra training.

d) Expected correction time: around 2–3 seconds per input.

e) The tool can help students improve writing and grammar skills interactively.

7. Conclusion

In summary, this project successfully integrates **machine learning, NLP, and user-centered design** to deliver a tool that is **innovative, reliable, and educationally valuable**, serving as a strong example of how AI can be leveraged to support learning and improve language skills in an accessible and interactive way.

References

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5. Hugging Face Documentation: https://huggingface.co/transformers
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